Unit 3 Progress Check: MCQ Part B

1. Consider the following class definition.

```
public class MyClass
{
  private int x;

  public MyClass(int start)
  {
    x = start;
  }

  public void updateX(int amount)
  {
    x += amount;
  }
}
```

Suppose that tester is a properly instantiated reference to a MyClass object and num is an int value. Which of the following best describes the conditions under which the value of the instance variable x is unchanged as a result of the call tester.updateX(num)?

(A) When num is 0 ✓

(B) When num is 1

(C) When num is positive

(D) When num is negative

Answer A

Correct. The updateX method adds amount to the value of the instance variable x and stores the result in x. When num is 0, the value passed to amount is 0, and x is increased by 0. Therefore, the value of x does not change.

Unit 3 Progress Check: MCQ Part B

2. Consider the following class definition.

```
public class XYPoint
{
  private int x;
  private int y;

  public XYPoint(int xVal, int yVal)
  {
    x = xVal;
    y = yVal;
  }

  public String getPoint()
  {
    /* missing code */
  }
}
```

The following code segment appears in a class other than XYPoint.

```
XYPoint p1 = new XYPoint(3, -2);
XYPoint p2 = new XYPoint(4, 0);
System.out.println(p1.getPoint());
System.out.println(p2.getPoint());
```

This code segment is intended to produce the following output.

```
(3, -2)
(4, 0)
```

Which of the following statements can be used to replace /* missing code */ so that this code segment produces the intended output?

Unit 3 Progress Check: MCQ Part B

(A)

```
System.out.println("(x, y)");
```

(B)

```
System.out.println("(" + x + ", " + y + ")");
```

(C)

```
return "(x, y)";
```

(D) ✓

```
return "(" + x + ", " + y + ")";
```

Answer D

Correct. This code segment builds and returns a correctly formatted string using the values of the instance variables x and y.

Unit 3 Progress Check: MCQ Part B

3. Consider the following class definition.

```
public class Fraction
{
  private int numerator;
  private int denominator;

  public Fraction(int n, int d)
  {
    numerator = n;
    denominator = d;
  }

  double getDecimal()
  {
    return (double) numerator / denominator;
  }
}
```

Suppose that fr is a properly instantiated reference to a Fraction object. Which of the following best describes the conditions under which the call fr.getDecimal() will fail to return a value?

(A) When the instance variables numerator and denominator have the same value

(B) When the instance variables numerator and denominator have different values

(C) When the instance variable denominator is equal to 0 ✓

(D) When the instance variable denominator is negative

Answer C

Correct. The getFraction method returns the double quotient when numerator is divided by denominator. When denominator is 0, an ArithmeticException occurs. In all other cases, a double value is returned.

Unit 3 Progress Check: MCQ Part B

4. Consider the following method.

```
public String mystery(String message, int x)
{
  message = message.substring(x);
  return message;
}
```

The following code segment appears in a method in the same class as mystery.

```
String str1 = "abcde";
String str2 = mystery(str1, 3);
```

Which of the following best describes the behavior of this code segment?

(A) str1 is modified by the call to mystery; str1 and str2 are equal in value after executing the code segment.

(B) str1 is modified by the call to mystery; str1 and str2 have differing values after executing the code segment.

(C) str1 is not modified by the call to mystery; str1 and str2 are equal in value after executing the code segment.

(D) str1 is not modified by the call to mystery; str1 and str2 have differing values after executing the code segment. ✓

Answer D

Correct. Since the String argument passed into the method is not mutable, it is not modified by a call to mystery. The string that is returned by the method is a substring of the string passed into the method, so it will have a different value.

Unit 3 Progress Check: MCQ Part B

5. Consider the following class definition.

```
public class Location
{
  private int row, col;

  public Location()
  { /* implementation not shown */ }

  /* There may be other constructors
     and methods not shown. */
}
```

The following method is intended to create a copy of a Location object.

```
public Location clone(Location loc)
{
  Location loc1 = new Location();
  loc1.row = loc.row;
  loc1.col = loc.col;
  return loc1;
}
```

Which of the following best describes the conditions under which the method works as intended?

(A) The method will run correctly only if it is defined in a class other than Location.

(B) The method will run correctly only if it is defined inside the Location class. ✓

(C) The method will always run correctly.

(D) The method will never run correctly.

Answer B

Correct. The method accesses private instance variables so it must be defined inside the Location class.

Unit 3 Progress Check: MCQ Part B

6. Consider the following class definition.

```
public class Item
{
  // maintains the price of an item
  private int price;

  public int getPrice()
  {
    return price;
  }

  /* There may be instance variables, constructors,
     and other methods not shown. */
}
```

The following method appears in a class other than Item. The method is intended to calculate and return the sales tax of the item specified by the parameter myItem. The sales tax will be calculated as the cost of the item times the value of the parameter taxRate.

```
public double getTax(Item myItem, double taxRate)
{
  double cost = /* missing code */ ;
  return cost * taxRate;
}
```

Which of the following expressions can be used to replace /* missing code */ so that this method works as intended?

(A)

Item.price

(B)

myItem.price

(C)

Item.getPrice()

(D) ✓

myItem.getPrice()

Unit 3 Progress Check: MCQ Part B

Answer D

Correct. The accessor method can be called on an Item object.

Unit 3 Progress Check: MCQ Part B

7. Consider the following StringFinder class.

```
public class StringFinder
{
  private String fullString;
  private static String target = "x";

  public StringFinder(String fs)
  {
    fullString = fs;
  }

  public static void updateTarget(String newTarget)
  {
    target = newTarget;
  }

  public int targetLocation()
  {
    int temp = fullString.indexOf(target);
    target = "z";
    return temp;
  }
}
```

The following code segment appears in a class other than StringFinder.

```
StringFinder music = new StringFinder("jazz");
StringFinder fire = new StringFinder("blaze");
StringFinder animal = new StringFinder("zebra");
int musicLoc = music.targetLocation();
StringFinder.updateTarget("a");
int fireLoc = fire.targetLocation();
int animalLoc = animal.targetLocation();
System.out.println(musicLoc + " " + fireLoc + " " + animalLoc);
```

What is printed as a result of executing this code segment?

Unit 3 Progress Check: MCQ Part B

(A)

-1 -1 -1

(B) ✓

-1 2 0

(C)

-1 2 4

(D)

0 2 4

Answer B

Correct. The first call to targetLocation returns -1 since the value of target is "x" and "x" is not found in "jazz", which is the value of fullString for the music object. The next statement calls the class method updateTarget and changes the value of the class variable target to "a". The second call to targetLocation returns 2 since the value of target is "a" and "a" is at index 2 in "blaze", which is the value of fullString for the fire object. The value of target is changed to "z" in the method. The third call to targetLocation returns 0 since the value of target is "z" and "z" is at index 0 in "zebra", which is the value of fullString for the animal object.

Unit 3 Progress Check: MCQ Part B

8. Consider the following class definition.

```
public class Widget
{
  private int number;
  private static String word = "start";

  public Widget()
  { /* implementation not shown */ }
}
```

The following code segment appears in a class other than Widget.

```
int result = Widget.doSomething();
```

Which of the following implementations of doSomething will allow this code segment to run without error when added to the Widget class?

Unit 3 Progress Check: MCQ Part B

(A)

```
public int doSomething()
{
  return number;
}
```

(B)

```
public int doSomething()
{
  return word.length();
}
```

(C)

```
public static int doSomething()
{
  return number;
}
```

(D) ✓

```
public static int doSomething()
{
  return word.length();
}
```

Answer D

Correct. The calling statement uses the Widget class name, so the method must be designated static. Class methods

Unit 3 Progress Check: MCQ Part B

can access only the values of class variables, so only the variable word can be referenced.

Unit 3 Progress Check: MCQ Part B

9. Consider the following class declaration.

```
public class TestObject
{
  private double var1;
  private static int var2 = 0;

  public TestObject(double p)
  {
    var1 = p;
    var2++;
  }

  public void printTestObject()
  {
    System.out.println(var1 + ", " + var2);
  }
}
```

The following code segment appears in a class other than TestObject. Assume that no other TestObject objects have been created.

```
TestObject obj1 = new TestObject(2.5);
TestObject obj2 = new TestObject(10.2);
obj1.printTestObject();
```

What is printed as a result of executing this code segment?

(A)

2.5, 0

(B)

2.5, 1

(C) ✓

2.5, 2

(D)

10.2, 2

Unit 3 Progress Check: MCQ Part B

Answer C

Correct. The instance variable var1 has the value of 2.5 for obj1, and the class variable var2 has the value 2. var2 was incremented by 1 when obj1 was instantiated and then incremented by 1 again when obj2 was instantiated.

Unit 3 Progress Check: MCQ Part B

10. Consider the following class definition.

```
public class ComputeObject
{
  private int limit;
  private int val;

  public ComputeObject()
  {
    limit = 7;
    val = 10;
  }

  public int sumProd(int limit)
  {
    int total = 0;
    for (int val = 0; val < limit; val++)
    {
      total += val;
    }
    total *= val;
    return total;
  }
}
```

The following code segment appears in a class other than ComputeObject.

```
ComputeObject s = new ComputeObject();
System.out.println(s.sumProd(5));
```

Which of the following best describes the behavior of this code segment?

(A) The sumProd method accumulates the sum of the integers 0 through 4, multiplies the sum by 5, and returns the value 50, which is then printed.

(B) The sumProd method accumulates the sum of the integers 0 through 4, multiplies the sum by 10, and returns the value 100, which is then printed. ✓

(C) The sumProd method accumulates the sum of the integers 0 through 6, multiplies the sum by 7, and returns the value 147, which is then printed.

(D) The sumProd method accumulates the sum of the integers 0 through 6, multiplies the sum by 10, and returns the value 210, which is then printed.

Answer B

Unit 3 Progress Check: MCQ Part B

Correct. The given code uses the value of the sumProd parameter limit (5) to iterate 5 times. Inside the loop, the variable val is incremented from 0 to 4, accumulating a sum of 10. The loop terminates when val is incremented to 5. Once the for loop ends, the loop variable val is out of scope. The multiplication statement uses the value of the instance variable val (10), resulting in 100.

Unit 3 Progress Check: MCQ Part B

11. Consider the following class definition.

```
public class Stars
{
  private int size;
  private int width;

  public Stars(int n, int w)
  {
    size = n;
    width = w;
  }

  public void draw(int size)
  {
    for (int i = 1; i <= size; i++)
    {
      line(i, width - 1);
    }
  }

  public void line(int n, int width)
  {
    for (int i = 0; i < n; i++)
    {
      for (int j = 0; j < width; j++)
      {
        System.out.print("*");
      }
    }
    System.out.println();
  }
}
```

The following code segment appears in a class other than Stars.

```
Stars s = new Stars(3, 2);
s.draw(5);
```

What is printed as a result of executing this code segment?

Unit 3 Progress Check: MCQ Part B

(A)

```
*
**
***
```

(B)

```
**
****
******
```

(C) ✓

```
*
**
***
****
*****
```

(D)

```
**
****
******
********
**********
```

Answer C

Correct. The constructor creates an object s with instance variable size = 3 and width = 2. The draw method is called and a local variable named size is created with size = 5. This local variable is used in place of the instance variable

Unit 3 Progress Check: MCQ Part B

size throughout the draw method. The draw method iterates 5 times with loop variable i taking on values 1 through 5. Each iteration of the loop calls line with i for parameter n and 1 for parameter width. This local variable width is used in place of the instance variable width throughout the line method. The line method prints n * width asterisks horizontally during each call.

12. Consider the following class, which models a bank account. The deposit method is intended to update the account balance by a given amount; however, it does not work as intended.

```
public class BankAccount
{
  private String accountOwnerName;
  private double balance;
  private int accountNumber;

  public BankAccount(String name, double initialBalance, int acctNum)
  {
    accountOwnerName = name;
    balance = initialBalance;
    accountNumber = acctNum;
  }

  public void deposit(double amount)
  {
    double balance = balance + amount;
  }
}
```

Which of the following best explains why the deposit method does not work as intended?

(A) The deposit method must have a return statement.

(B) In the deposit method, the variable balance should be replaced by the variable initialBalance.

(C) In the deposit method, the variable balance is declared as a local variable and is different from the instance variable balance. ✓

(D) The variable balance must be passed to the deposit method.

Answer C

Correct. There are two double variables with the same name, balance, one of them declared as an instance variable in the BankAccount class and the other one declared as a local variable inside the deposit method.

Unit 3 Progress Check: MCQ Part B

13. Consider the following class definition.

```
public class Player
{
  private int base;
  private int bonus;

  public Player()
  { /* implementation not shown */ }

  public int getTotal()
  {
    return base + bonus;
  }

  public boolean hasHigherScore(Player other)
  {
    /* missing code */
  }
}
```

The Player class contains a getTotal method, which returns the total score for the player.

The class also contains a hasHigherScore method, which has a Player object as a parameter. The hasHigherScore method, when called on a given Player object, is intended to return true if the total score of the given Player is greater than the total score of the Player specified by the parameter. It is intended to return false otherwise.

For example, suppose p1 and p2 are valid Player objects. If p1 has a higher total score than p2, p1.hasHigherScore(p2) should return true.

Which of the following can be used to replace /* missing code */ so that the hasHigherScore method works as intended?

Unit 3 Progress Check: MCQ Part B

(A)

```
return this > other;
```

(B)

```
return getTotal(this) > getTotal(other);
```

(C)

```
return this.getTotal() > getTotal();
```

(D) ✓

```
return this.getTotal() > other.getTotal();
```

Answer D

Correct. The keyword this refers to the current object (the Player whose hasHigherScore method is being called). The call this.getTotal() will return the total score of the given Player, and the call other.getTotal() will return the total score of the other Player.

Unit 3 Progress Check: MCQ Part B

14. Consider the following class declaration.

```
public class Thing
{
  private int val;

  public Thing(int v)
  {
    val = v;
  }

  public int getVal()
  {
    return val;
  }

  public String mystery(Thing other)
  {
    if (this == other)
    {
      return "yes";
    }
    else if (this.val == other.getVal())
    {
      return "maybe";
    }
    else return "no";
  }
}
```

The following code segment appears in a class other than Thing.

```
Thing apple = new Thing(5);
Thing banana = new Thing(5);

System.out.println(apple.mystery(banana));
System.out.println(banana.mystery(banana));
```

What, if anything, is printed as a result of executing this code segment?

Unit 3 Progress Check: MCQ Part B

(A)

yes

yes

(B) ✓

maybe

yes

(C)

maybe

maybe

(D) Nothing is printed because the Thing class does not compile.

Answer B

Correct. The keyword this refers to the current object (the Thing whose mystery method is being called). For the call apple.mystery(banana), the expression this == other evaluates to false because apple and banana refer to different objects. The expression this.val == other.getVal() evaluates to true because apple's instance variable val is equal to banana's instance variable val. Therefore, the call apple.mystery(banana) returns "maybe". For the call banana.mystery(banana), the expression this == other evaluates to true because this and other refer to the same object (banana). Therefore, the call banana.mystery(banana) returns "yes".

Unit 3 Progress Check: MCQ Part B

15. Consider the following class definition.

```
public class Friend
{
  private String name;     // Line 3

  public Friend(String name)
  {
    name = name;          // Line 7
  }

  public String getName()
  {
    return name;          // Line 12
  }
}
```

The following code segment appears in a class other than Friend. It is intended to print the string "Jessie" but does not work as intended because of an error in the Friend class.

```
Friend bestie = new Friend("Jessie");
System.out.println(bestie.getName());
```

Which of the following changes can be made to the Friend class so that this code segment works as intended?

(A) Changing line 3 to private String this.name;

(B) Changing line 7 to this.name = name; ✓

(C) Changing line 7 to name = this.name;

(D) Changing line 12 to return this.name;

Answer B

Correct. The keyword this refers to the current object. In the given class declaration, the parameter in the constructor has the same name as the instance variable. That means that the statement name = name; assigns the value of the parameter to itself. The statement this.name = name; will assign the value of the parameter name to the instance variable name associated with this Friend object.